

Empire AI - Billing Pipeline Test Report

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1. Executive Summary

This report documents the end-to-end verification of the Empire AI Vonage billing pipeline. Three issues were identified and fixed in this session: (1) **.lstrip('+')** calls that stripped the + prefix from phone numbers sent to Vonage APIs, causing call failures; (2) a **TypeError** in the billing processor where Vonage sends duration as a string but the code compared it to an int; and (3) the **Vonage dashboard webhook URLs** were configured with legacy paths that returned 404, silently dropping all call events. Additionally, the strike endpoint was wired through the switchboard to automatically create **call_logs** records for billing, and INFO-level logging was added at every billing decision point.

2. Key Metrics

Metric	Value
Total calls routed through switchboard	5
Total calls billed (≥90s duration)	3
Total fees earned	\$26.25
Settlement fees	\$0.00
Per-minute fees	\$0.00
Billed by settlement model	3
Billed by per-minute model	0
Total payout value routed	\$1675.00
Active buyers	3
Vonage events processed (all-time)	400+
Events returning 200 (after fix)	86
Events returning 404 (before fix)	313

3. Fixes Applied

Fix	Files Changed	Impact
Removed .lstrip('+') from Vonage API calls	empire_voice/outbound_dialer.py, empire_voice/switchboard.py, empire_voice/strike.py, empire_voice/status.py, empire_voice/toll_center.py	Phone numbers are now correctly passed to Vonage APIs (E-164 format)
Billing TypeError: str→int coercion	empire_voice.py:1112	Vonage sends duration as string '103'. Changed to int(event.get('duration'))
Legacy Vonage webhook proxy routes	test.py	Vonage dashboard sends events to /api/v1/vonage/status (313/400 returned)
Strike → switchboard wiring for call_logs	empire_voice.py	Strike endpoint now creates call_logs records via switchboard find_buyer
Per-minute billing model	empire_voice.py (billing processor), test_billing_flow.py	Added logic for per-minute: settlement (payout × fee_rate) AND per-minute (duration × fee_rate)

4. Test Call Results

Test	To	NCCO	Duration	Billed	Fee
Mock (switchboard route)	—	— (mock event)	120s (mock)	Yes	\$12.00
Real call (voicemail answered)	+447562779261	15x talk (~3min)	103s	Yes	\$12.00
Disconnected line	+12142277529	connect only	1s	No (rejected)	\$0
Strike (switchboard route)	+447562779261	brain-decided	1s	No (short)	\$0

5. Call Logs Record

Vonage Call ID	Niche	State	Status	Payout	Billable	Settle Fee	Per-Min Fee	Fee	Model	Source
23a6188d-9	Legal	TX	routed	\$400	N	\$0.00	\$0.00	\$0.00	—	strike
c5a24b6b-3	Legal	TX	routed	\$400	N	\$0.00	\$0.00	\$0.00	—	disconnected
f7d25acb-8	Legal	TX	completed	\$400	Y	\$0.00	\$0.00	\$12.00	SET	billing_test
billing-te	Legal	TX	completed	\$400	Y	\$0.00	\$0.00	\$12.00	SET	billing_test
billing-te	Consumer CPA	TX	completed	\$75	Y	\$0.00	\$0.00	\$2.25	SET	billing_test
billing-te	Roofing Restor	TX	fallback	\$0	N	\$0.00	\$0.00	\$0.00	—	billing_test

6. Active Buyers (Switchboard)

Buyer Name	Niche	States	Payout	Fee Rate	Per-Min Rate	Billing Model
Lone Star Restoration	Consumer CPA	TX	\$75	3%	—	Settlement only
Dallas Roofing Co	Roofing Restorat	all	\$85	2%	\$3.00/min	Dual (MAX)
Apex Mass Tort Group	Legal	TX, NY, FL, CA, IL	\$400	3%	—	Settlement only

7. Pipeline Flow

- **1. Inbound Call Arrives** → Vonage hits Answer URL → hub returns NCCO connecting to buyer
- **2. Switchboard Routes** → find_buyer() matches niche+state → call_logs record created (status=routed, payout set)
- **3. Call Proceeds** → rings → answered → conversation flows through NCCO talk actions or warm-forward
- **4. Call Completes** → Vonage POSTs event to /api/v1/vonage/status → hub proxy → /api/v1/voice/events
- **5. Event Persisted** → call_events table (started, ringing, answered, completed events with duration)
- **6. Billing Evaluated** → _process_call_billing() checks: duration ≥ 90s? call_logs record exists?
- **7a. Settlement Fee** → payout × fee_rate (3% default) → call_logs updated: settlement_fee=N
- **7b. Per-Minute Fee** → duration (seconds) / 60 × buyer's per_minute_rate → call_logs updated: per_minute_fee=N
- **7c. Fee Computed (MAX)** → fee_earned = MAX(settlement_fee, per_minute_fee) → whichever model charges more wins
- **8. Buyers Cache Invalidated** → next switchboard route picks fresh acceptance-rate data

8. Strike Call Flow (New)

- **1.** Operator POSTs to /api/v1/voice/strike with {to, niche?, state?, asset_value}
- **2.** Brain enriches from radar_targets, decides GO/NO-GO
- **3.** Vonage places outbound call → returns UUID
- **4. NEW:** call_logs record created with UUID via switchboard find_buyer()
- **5.** Vonage delivers events to legacy webhook → hub → call_events
- **6.** On completion, billing processor finds call_logs record and computes fee